The probabilistic interpretation has the virtue that if an electron wave does what other waves can do—for instance, slam into some obstacle and develop all sorts of distinct ripples—it does not mean that the electron itself has shattered into separate pieces. Rather, it means that there are now a number of locations where the electron might be found with a non-negligible probability. In practice this means that if a particular experiment involving an electron is repeated over and over again in an absolutely identical manner, the same answer for, say, the measured position of an electron will not be found over and over again. Rather, the subsequent repeats of the experiment will yield a variety of different results with the property that the number of times the electron is found at any given location is governed by the shape of the electron's probability wave. If the probability wave (more precisely, the square of the probability wave) is twice as large at location A than at location B, then the theory predicts that in a sequence of many repeats of the experiment the electron will be found at location A twice as often as at location B. Exact outcomes of experiments cannot be predicted; the best we can do is predict the probability that any given outcome may occur.

Even so, as long as we can determine mathematically the precise form of probability waves, their probabilistic predictions can be tested by repeating a given experiment numerous times, thereby experimentally measuring the likelihood of getting one particular result or another. Just a few months after de Broglie's suggestion, Schrödinger took the decisive step toward this end by determining an equation that governs the shape and the evolution of probability waves, or as they came to be known, wave functions. It was not long before Schrödinger's equation and the probabilistic interpretation were being used to make wonderfully accurate predictions. By 1927, therefore, classical innocence had been lost. Gone were the days of a clockwork universe whose individual constituents were set in motion at some moment in the past and obediently fulfilled their inescapable, uniquely determined destiny. According to quantum mechanics, the universe evolves according to a rigorous and precise mathematical formalism, but this framework determines only the probability that any particular future will happen—not which future actually ensues.